Clément Omnès

contact@clementomnes.dev +33 X XX XX XX XX French citizen Graduating Oct 2028
github.com/clementomnes linkedin.com/in/clementomnes clementomnes.dev Available May 3 – Oct 15, 2027 (17+ weeks)

Software engineering student with production experience as an apprentice and intern, seeking a research internship in distributed systems and concurrency, with an experimental approach: implementing competing designs, measuring them, and turning prototypes into tested code.

Skills

Programming: Java, Python, TypeScript, JavaScript, C#, SQL, HTML, CSS

Technologies: Spring, React, Next.js, gRPC, RabbitMQ, WebSocket, PostgreSQL, MongoDB, Git, Claude Code (agentic AI coding)

Infrastructure: Oracle Cloud Infrastructure, Docker, Linux, Cloudflare Zero Trust, GitHub Actions CI/CD

Education

CESI Engineering School — Rouen, France

Master of Science in Computer Science and Engineering – CGPA: 3.8 / 4.0 — *Oct 2025 – Oct 2028 (Expected)*

Relevant coursework: Distributed Application Development, Design Patterns, Software Architecture & Engineering, System Programming (.NET), Software Modelling & Versioning, SI Administration & Security

Applied projects: designed and implemented distributed systems covering multithreading, non-synchronised applications, and information system security

Rouen University Institute of Technology — Rouen, France

Bachelor of Science in Multimedia and Internet Technologies – CGPA: 3.69 / 4.0 — *Sep 2022 – Jul 2025*

Relevant coursework: Advanced Front-End & Back-End Development, Web Deployment & Hosting, Interactive Devices, UX & Experience Design

Experience

R3mScore — Rouen, France

Apprentice Software Engineer — *Sep 2024 – Present*

- Designed and owned end-to-end a **Word-to-survey import/export engine** parsing .docx questionnaires into structured surveys (conditional logic, loops, adaptive questions), replacing **20min–3h of manual setup per survey**, unique among competitors.
- Eliminated type-related runtime errors across a **15,000+ line** production codebase by migrating the frontend from JavaScript to TypeScript incrementally, with **no feature freeze and no regression in production**.
- Raised test coverage across critical features, reducing production incidents, by implementing JUnit and Vitest suites for backend and frontend.
- Cut frontend build time **10× (20 min → 2 min)**, boosting deployment frequency, by migrating the build pipeline to Vite.
- Shipped customer-facing features used daily, reducing manual operations, by building end-to-end Java/Spring and React.js modules from requirements to production.
- Built an **in-house LLM orchestration layer** (lighter LangChain4j-style clone) over OpenAI and Mistral in Java, enabling streaming conversations, follow-up generation, and multilingual prompts with JSON Schema validation.

Software Engineer Intern — *Apr 2024 – Jun 2024*

- Simplified large-scale survey workflows by implementing Word import/export pipelines for questionnaires and translations, automatic REF renaming and validation, and safer handling of quotas, weights, and URL parameters in data exports.
- Improved maintainability and code quality by introducing ESLint, adding unit and integration tests on collectors and reporting logic, and refactoring legacy React components and Java services to clearer, typed and tested modules.
- Boosted team productivity by creating internal AI tools that simplified the process of organizing and analyzing open-ended survey responses including a prompt editor for refining AI queries.

Projects

Breezy - Full-stack social platform (microservices monorepo) — TypeScript, Node.js, gRPC, RabbitMQ, Next.js, Docker

- **Tech lead on a team of 4**, architecting an 8-service microservices platform with per-service databases and gRPC + RabbitMQ communication behind an Nginx gateway.
- Built **real-time messaging over WebSocket** (1-to-1 and group, read receipts) and live notifications via SSE and Web Push, plus video streaming with HTTP range requests.
- Instrumented full observability with Prometheus, Grafana, and OpenTelemetry/Jaeger, and set up CI via GitHub Actions with an aggregated OpenAPI spec across services.

TSP Resolver - Metaheuristics for a constrained Traveling Salesman Problem — Python, NumPy, Pandas, NetworkX

- Implemented **Tabu Search** (standard and 2-opt) and **multi-start Simulated Annealing** in a team of 5, for a TSP variant with forbidden nodes, precedence constraints and toll costs.
- Built the **statistical analysis framework** comparing the team's metaheuristics on instances from **5 to 3,000 nodes** over multiple random seeds, showing that only 2-opt Tabu Search and multi-start Simulated Annealing improved their optimality gap as instances grew.

Languages & Interests

Languages: French (native), English (professional working proficiency)

Interests: Kung Fu, Music (Kora & trumpet player), Hiking